

If fuel tanks of simple geometry are used, the tank volume can be calculated directly. Wing-box fuel volume can be approximated by assuming a tapered box shape. For complex integral and bladder tanks, the tank volume is determined using a fuel-volume plot as shown in Fig. 10.23. This is constructed by measuring the cross-sectional area of the tanks at various fuselage locations, then plotting those cross-sectional areas on a volume plot similar to the aircraft volume plot previously discussed.

If a discrete tank is used, the actually available internal volume can be calculated by subtracting the wall thickness from the external dimensions. For integral and bladder tanks, the available tank volume must be reduced from the measured value to allow for wall thickness, internal structure, and bladder thickness.

A rule of thumb is to assume that 85% of the volume measured to the external skin surface is usable for integral wing tanks, and 92% is usable for integral fuselage tanks. If bladder tanks are used, the values become 77% for wing tanks and 83% for fuselage tanks.

Note in Fig. 10.23 that the fuel volume plot allows the estimation of the center of gravity (c.g.) for each fuel tank, which is the centroid of the area plotted for the tank. The total fuel c.g. is simply the weighted average of the individual tank c.g.'s, and should be close to the aircraft c.g.

LANDING GEAR AND SUBSYSTEMS

11.1 INTRODUCTION

Of the many internal components that must be defined in an aircraft layout, the landing gear will usually cause the most trouble. Landing gear must be placed in the correct down position for landing, and must somehow retract into the aircraft without chopping up the structure, obliterating the fuel tanks, or bulging out into the slipstream. This chapter covers landing-gear design as well as installation of other subsystems.

11.2 LANDING GEAR ARRANGEMENTS

The common options for landing-gear arrangement are shown in Fig. 11.1. The single main gear is used for many sailplanes because of its simplicity. The wheel can be forward of the center of gravity (c.g.), as shown here, or can be aft of the c.g. with a skid under the cockpit.

“Bicycle” gear has two main wheels, fore and aft of the c.g., with small “outrigger” wheels on the wings to prevent the aircraft from tipping sideways. The bicycle landing gear has the aft wheel so far behind the c.g. that the aircraft must takeoff and land in a flat attitude, which limits this type of gear to aircraft with high lift at low angles of attack (i.e., high-aspect-ratio wings with large camber and/or flaps). Bicycle gear has been used mainly on aircraft with narrow fuselage and wide wing span such as the B-47.

The “taildragger” landing gear has two main wheels forward of the c.g. and an auxiliary wheel at the tail. Taildragger gear is also called “conventional” landing gear, because it was the most widely used arrangement during the first 40 years of aviation. Taildragger gear provides more propeller clearance, has less drag and weight, and allows the wing to generate more lift for rough-field operation than does tricycle gear.

However, taildragger landing gear is inherently unstable. If the aircraft starts to turn, the location of the c.g. behind the main gear causes the turn to get tighter until a “ground loop” is encountered, and the aircraft either drags a wingtip, collapses the landing gear, or runs off the side of the runway. To prevent this the pilot of a taildragger aircraft must align the aircraft almost perfectly with the runway at touchdown, and “dance” on the rudder pedals until the aircraft stops.

The most commonly used arrangement today is the “tricycle” gear, with two main wheels aft of the c.g. and an auxiliary wheel forward of the c.g.. With a tricycle landing gear, the c.g. is ahead of the main wheels so the aircraft is stable on the ground and can be landed at a fairly large “crab”

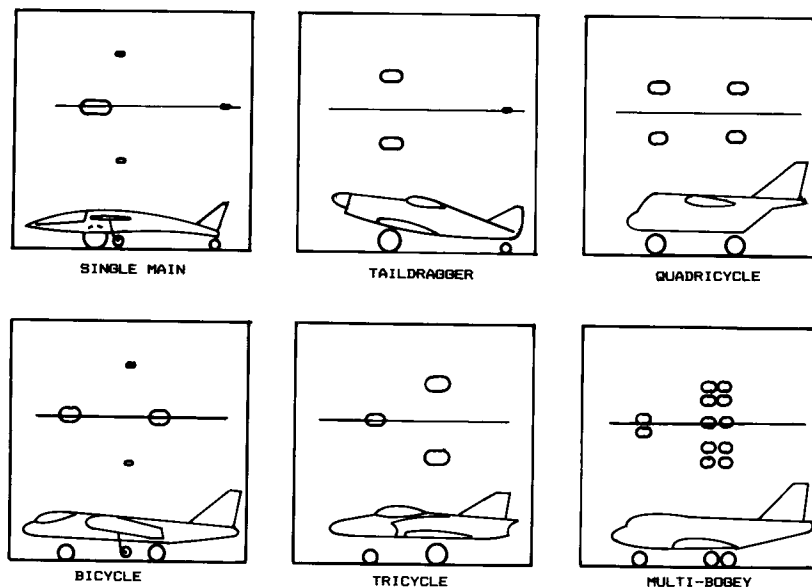


Fig. 11.1 Landing gear arrangements.

angle (i.e., nose not aligned with the runway). Also, tricycle landing gear improves forward visibility on the ground and permits a flat cabin floor for passenger and cargo loading.

Quadricycle gear is much like bicycle gear but with wheels at the sides of the fuselage. Quadricycle gear also requires a flat takeoff and landing attitude. It is used on the B-52 and several cargo planes where it has the advantage of permitting a cargo floor very low to the ground.

The gear arrangements described above are also seen with two, four, or more wheels in place of the single wheels shown in Fig. 11.1. As aircraft weights become larger, the required wheel size for a single wheel capable of holding the aircraft's weight becomes too large. Then multiple wheels are used to share the load between reasonably-sized tires.

Also, it is very common to use twin nose-wheels to retain some control in the event of a nose-wheel flat tire. Similarly, multiple main wheels (i.e., total of four or more) are desirable for safety. When multiple wheels are used in tandem, they are attached to a structural element called a "bogey," or "truck," which is attached to the end of the shock-absorber strut.

Typically an aircraft weighing under about 50,000 lb will use a single main wheel per strut, although for safety in the event of a flat tire it is always better to use two wheels per strut. Between 50,000 and 150,000 lb, two wheels per strut are typical. Two wheels per strut are sometimes used for aircraft weighing up to about 250,000 lb.

Between aircraft weights of about 200,000 and 400,000 lb the four-wheel bogey is usually employed; for aircraft over 400,000 lb, four bogeys, each with four or six wheels, spread the total aircraft load across the runway pavement.

Except for light aircraft and a few fighters, most aircraft use twin nose-wheels to retain control in the event of a flat nose tire. Carrier-based aircraft must use twin nose-wheels at least 19 inches in diam to straddle the catapult-launching mechanism. The massive C-5 employs four nose-wheels to spread the tire load, permitting operation off of relatively soft fields.

Guidelines for layout of a bicycle landing gear are shown in Fig. 11.2. The c.g. should be aft of the midpoint between the two wheels.

The requirements for taildragger gear are shown in Fig. 11.3. The tail-down angle should be about 10–15 deg with the gear in the static position (i.e., tires and shock absorbers compressed the amount seen when the aircraft is stationary on the ground at takeoff gross weight).

The c.g. (most forward and most aft) should fall between 16–25 deg back from vertical measured from the main wheel location. If the c.g. is too far forward the aircraft will tend to nose over, and if it is too far back it will tend to groundloop.

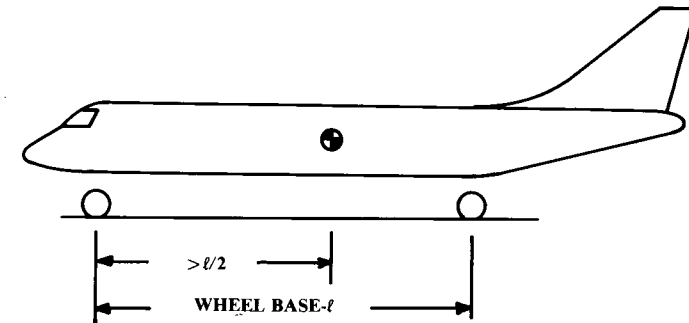


Fig. 11.2 Bicycle landing gear.

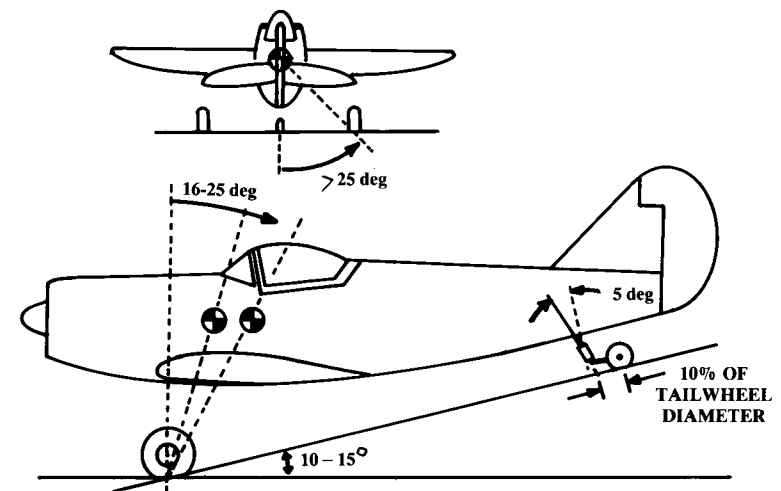


Fig. 11.3 Taildragger landing gear.

To prevent the aircraft from overturning the main wheels should be laterally separated beyond a 25 deg angle off the c.g., as measured from the rear in a tail-down attitude.

The layout of tricycle landing gear as shown in Fig. 11.4 is even more complex. The length of the landing gear must be set so that the tail doesn't hit the ground on landing. This is measured from the wheel in the static position assuming an aircraft angle of attack for landing which gives 90% of the maximum lift. This ranges from about 10–15 deg for most types of aircraft.

The “tipback angle” is the maximum aircraft nose-up attitude with the tail touching the ground and the strut fully extended. To prevent the aircraft from tipping back on its tail, the angle off the vertical from the main wheel position to the c.g. should be greater than the tipback angle or 15 deg, whichever is larger.

For carrier-based aircraft this angle frequently exceeds 25 deg, implying that the c.g. for carrier-based aircraft is well forward of the main wheels. This insures that the rolling of the deck will not cause an aircraft to tip back on its tail.

However, this also makes it difficult to lift the nose for a runway takeoff. If the nose wheel is carrying over 20% of the aircraft's weight, the main gear is probably too far aft relative to the c.g.

On the other hand, if the nose wheel is carrying less than 5% of the aircraft's weight, there will not be enough nose-wheel traction to steer the aircraft. The optimum range for the percent of the aircraft's weight which

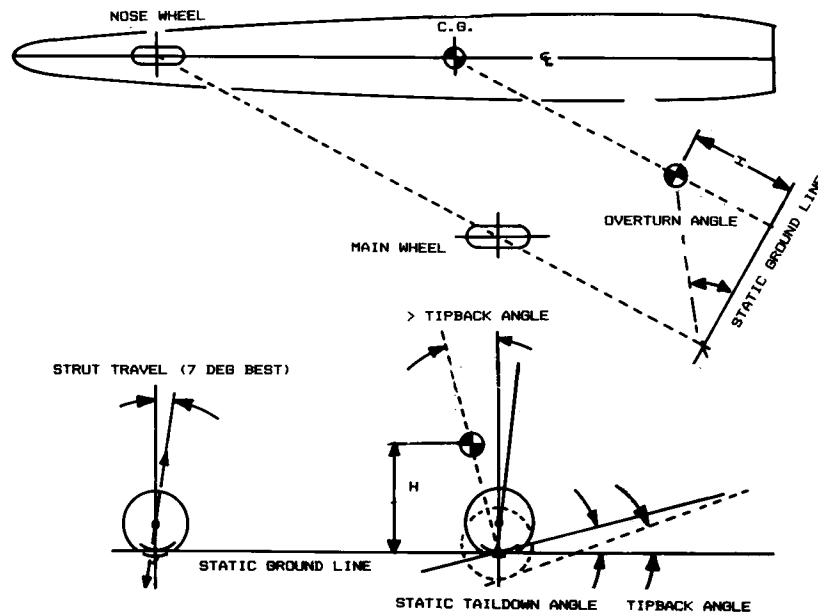


Fig. 11.4 Tricycle landing gear geometry.

is carried by the nose wheel is about 8–15%, for the most-aft and most-forward c.g. positions.

The “overturn angle” is a measure of the aircraft's tendency to overturn when taxied around a sharp corner. This is measured as the angle from the c.g. to the main wheel, seen from the rear at a location where the main wheel is aligned with the nose wheel. For most aircraft this angle should be no greater than 63 deg (54 deg for carrier-based aircraft).

Figure 11.4 also shows the desired strut-travel angle as about 7 deg. This optimal angle allows the tire to move upwards and backwards when a large bump is encountered, thus tending to smooth out the ride. However, any strut-travel angle from purely vertical to about 10 deg aft of vertical is acceptable. Strut geometry in which the tire must move forward as it moves up is undesirable.

11.3 TIRE SIZING

Strictly speaking, the “wheel” is the circular metal object upon which the rubber “tire” is mounted. The “brake” inside the wheel slows the aircraft by increasing the rolling friction. However, the term “wheel” is frequently used to mean the entire wheel/brake/tire assembly.

The tires are sized to carry the weight of the aircraft. Typically the main tires carry about 90% of the total aircraft weight. Nose tires carry only about 10% of the static load but experience higher dynamic loads during landing.

For early conceptual design, the engineer can copy the tire sizes of a similar design or use a statistical approach. Table 11.1 provides equations developed from data in Ref. 1 for rapidly estimating main tire sizes (assuming that the main tires carry about 90% of the aircraft weight).

These calculated values for diameter and width should be increased about 30% if the aircraft is to operate from rough unpaved runways.

Nose tires can be assumed to be about 60–100% the size of the main tires. The front tires of a bicycle or quadricycle-gear aircraft are usually the same size as the main tires. Taildragger aft tires are about a quarter to a third the size of the main tires.

For a finished design layout, the actual tires to be used must be selected from a manufacturer's catalog. This selection is usually based upon the smallest tire rated to carry the calculated static and dynamic loads.

Table 11.1 Statistical tire sizing

Main wheels diameter or width (in.) = $A W_W^B$	Diameter		Width	
	A	B	A	B
General aviation	1.51	0.349	0.7150	0.312
Business twin	2.69	0.251	1.170	0.216
Transport/bomber	1.63	0.315	0.1043	0.480
Jet fighter/trainer	1.59	0.302	0.0980	0.467

W_W = Weight on Wheel

Calculation of the static loads on the tires is illustrated in Fig. 11.5 and Eqs. (11.1–11.3). The additional dynamic load on the nose tires under a 10 fps per second braking deceleration is given in Eq. (11.4). Note that these loads are divided by the total number of main or nose tires to get the load per tire (wheel) “ W_w ,” which is used for tire selection.

$$(\text{Max Static Load}) = W \frac{N_a}{B}$$

(11.1)

$$(\text{Max Static Load})_{\text{nose}} = W \frac{M_f}{B}$$

(11.2)

$$(\text{Min Static Load})_{\text{nose}} = W \frac{M_a}{B}$$

(11.3)

$$(\text{Dynamic Braking Load})_{\text{nose}} = \frac{10HW}{gB}$$

(11.4)

Equation (11.4) assumes a braking coefficient (μ) of 0.3, which is typical for hard runways. This results in a deceleration of 10 ft/s².

To insure that the nose gear is not carrying too much or too little of the load, the parameter (M_a/B) should be greater than 0.05, and the parameter (M_f/B) should be less than 0.20 (0.08 and 0.15 preferred).

If an airplane is to be operated under FAR 25 provisions, a 7% margin should be added to all calculated wheel loads. Also, it is common to add an additional 25% to the loads to allow for later growth of the aircraft design.

Table 11.2 summarizes design data for typical tires (Ref. 30), including maximum load ratings, inflation pressures at that load, maximum landing

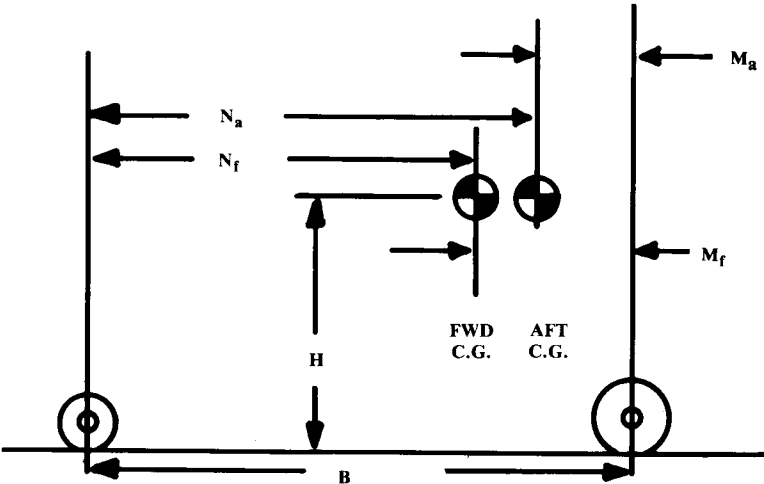


Fig. 11.5 Wheel load geometry.

Table 11.2 Tire data

Size	Speed, mph	Max load, lb	Infi, psi	Max width, in.	Max diam, in.	Rolling radius	Wheel diam	Number of plies
Type III								
5.00-4	120	1,200	55	5.05	13.25	5.2	4.0	6
5.00-4	120	2,200	95	5.05	13.25	5.2	4.0	12
7.00-8	120	2,400	46	7.30	20.85	8.3	8.0	6
8.50-10	120	3,250	41	9.05	26.30	10.4	10.0	6
8.50-10	120	4,400	55	8.70	25.65	10.2	10.0	8
9.50-16	160	9,250	90	9.70	33.35	13.9	16.0	10
12.50-16	160	12,800	75	12.75	38.45	15.6	16.0	12
20.00-20	174kt	46,500	125	20.10	56.00	22.1	20.0	26
Type VII								
16×4.4	210	1,100	55	4.45	16.00	6.9	8.0	4
18×4.4	174kt	2,100	100	4.45	17.90	7.9	10.0	6
18×4.4	217kt	4,350	225	4.45	17.90	7.9	10.0	12
24×5.5	174kt	11,500	355	5.75	24.15	10.6	14.0	16
30×7.7	230	16,500	270	7.85	29.40	12.7	16.0	18
36×11	217kt	26,000	235	11.50	35.10	14.7	16.0	24
40×14	174kt	33,500	200	14.00	39.80	16.5	16.0	28
46×16	225	48,000	245	16.00	45.25	19.0	20.0	32
50×18	225	41,770	155	17.50	49.50	20.4	20.0	26
Three Part Name								
18×4.25-10	210	2,300	100	4.70	18.25	7.9	10.0	6
21×7.25-10	210	5,150	135	7.20	21.25	9.0	10.0	10
28×9.00-12	156kt	16,650	235	8.85	27.60	11.6	12.0	22
37×14.0-14	225	25,000	160	14.0	37.0	15.1	14.0	24
47×18-18	195kt	43,700	175	17.9	46.9	19.2	18.0	30
52×20.5-23	235	63,700	195	20.5	52.0	21.3	23.0	30

speed, tire width, diameter, and “rolling radius” (i.e., radius when under load, typically two-thirds of tire radius). For a complete listing of available tires, a “tire book” can be obtained from the tire manufacturers.

The data in Table 11.2 is organized by tire type. A Type III tire, used for most piston-engined aircraft, has a wide tread and low internal pressure. The identifying numbers for a Type III tire, such as 8.50-10, refer to the approximate tire width (8.2–8.7 in.) and wheel rim diameter (10 in.). The tire outside diameter must be obtained from a tire book such as Ref. 30.

Type VII tires, used by most jet aircraft, operate under higher internal pressures, which reduces their size. Also, the Type VII tires are designed for higher landing speeds. They are identified by their approximate external

dimensions. For example, an 18×5.7 tire has an outside diameter of 17.25–17.8 in. and a width of 5.25–5.6 in. These actual numbers are obtained from a tire book.

For a while, the newest and highest-pressure tires were called “Type VIII,” but are now simply referred to as “new design” or “three-part name” tires. Designed for specific requirements, they are identified by outside diameter, width, and rim diameter. For example, a $36 \times 10.00-18$ tire has diameter ranging from 35.75–36.6 in., width from 9.75–10.3 in., and rim diameter of 18 in.

Tires are selected by finding the smallest tire that will carry the calculated maximum loads (W_w). For the nose tire the total dynamic load must be carried as well as the maximum static load.

However, a tire is permitted to carry more dynamic load than the rated static value found in Table 11.2 or a tire book. A Type III tire is permitted a dynamic load of 1.4 times the static value. A Type VII or a “new design” tire is allowed to carry 1.3 times the static value.

The total dynamic nosewheel load (static plus dynamic) should therefore be divided by 1.4 or 1.3, and then used to pick the nose tire size for dynamic load. Both static and dynamic loadings should be used to determine a minimum size of nose-wheel tire. Then the larger of the two should be selected.

As a tire ages it loses ability to withstand its own internal pressure. This causes it to swell in size by about 2 or 3% in diameter and 4% in width. This swelling should be allowed for in designing the wheel wells and retraction geometry, or the wheel pants for a fixed-gear aircraft.

A tire supports a load almost entirely by its internal pressure. The load-carrying ability of the sidewalls and tread can be ignored. The weight carried by the tire (W_w) is simply the inflation pressure (P) times the tire's contact area with the pavement (A_p , also called “footprint area”), as shown in Fig. 11.6 and defined in Eq. (11.5).

$$W_w = PA_p \quad (11.5)$$

$$A_p = 2.3 \sqrt{wd} \left(\frac{d}{2} - R_r \right) \quad (11.6)$$

Equation (11.6), from Ref. 31, relates the pavement contact area (A_p) to the tire width (w), diameter (d), and rolling radius (R_r). [Note: do not confuse tire width (w) with weight on the wheel (W_w)].

Usually a tire is kept at about the same rolling radius as given in the tire book even when being used for a lower-than-maximum load. From Eq. (11.5), the internal pressure must therefore be proportionally reduced when a tire is operated at a lower than maximum load.

Operating a tire at a lower internal pressure will greatly improve tire life. Roughly speaking, operating a tire at half of its maximum rated load (hence pressure) will increase the number of landings obtained from the tire by a factor of six. However, this requires a larger tire causing greater drag, weight, and wheel-well size.

Tire pressure should also be reduced (i.e., larger tires used) if the aircraft will operate from soft or rough runways. Actual determination of the tire

size for a particular soft-field landing requirement is very complex (see Ref. 32). As a rough estimate, tires should be sized to keep internal pressures below the values in Table 11.3 (Ref. 33) for the desired application.

Sometimes the diameter of the tires is set by the braking requirements. Aircraft brakes are similar to automobile disk brakes, and are usually placed inside the wheels in all but the smallest aircraft. The wheel typically has a rim diameter of about half the total diameter of the tire mounted on it. Wheel rim diameters are provided in Table 11.2 or a tire book.

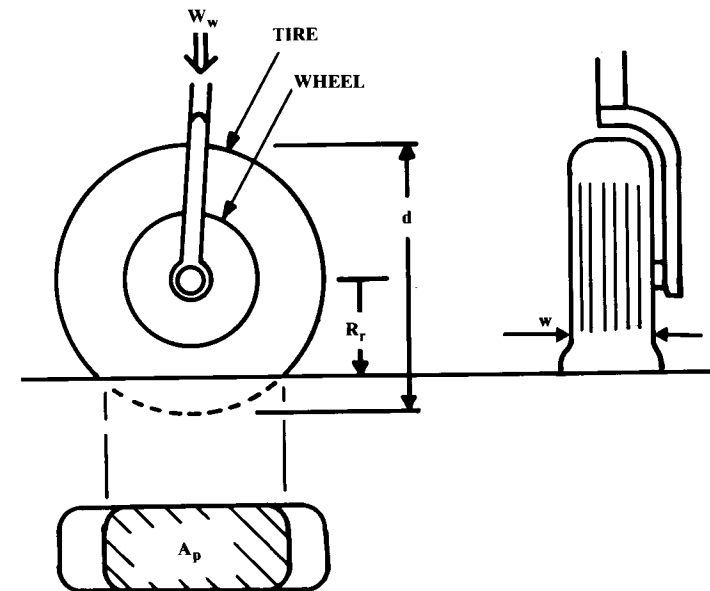


Fig. 11.6 Tire contact area.

Table 11.3 Recommended tire pressures

Surface	Maximum pressure, psi
Aircraft carrier	200 +
Major military airfield	200
Major civil airfield	120
Tarmac runway, good foundation	70–90
Tarmac runway, poor foundation	50–70
Temporary metal runway	50–70
Dry grass on hard soil	45–60
Wet grass on soft soil	30–45
Hard packed sand	40–60
Soft sand	25–35

The brakes must absorb the kinetic energy of the aircraft at touchdown, less the energy absorbed by aerodynamic drag and thrust reversing. These can be ignored by assuming that the brakes are applied when the aircraft has slowed to stall speed. This yields Eq. (11.7), which must be divided by the number of wheels with brakes to get the kinetic energy that must be absorbed by each brake. Note that while Western design practice puts brakes only on the main wheels, several Soviet designs also put brakes on the nose wheels.

$$KE_{\text{braking}} = \frac{1}{2} \frac{W_{\text{landing}}}{g} V_{\text{stall}}^2 \quad (11.7)$$

where $g = 32.2 \text{ ft/s}^2$.

The landing weight in Eq. (11.7) is not the same as the weight at the end of the design mission. To allow an emergency landing shortly after takeoff, the landing weight should be approximated as 80–100% of the takeoff weight.

A brake absorbs kinetic energy by turning it into heat. There is insufficient time during landing for much heat energy to be radiated to the air, so

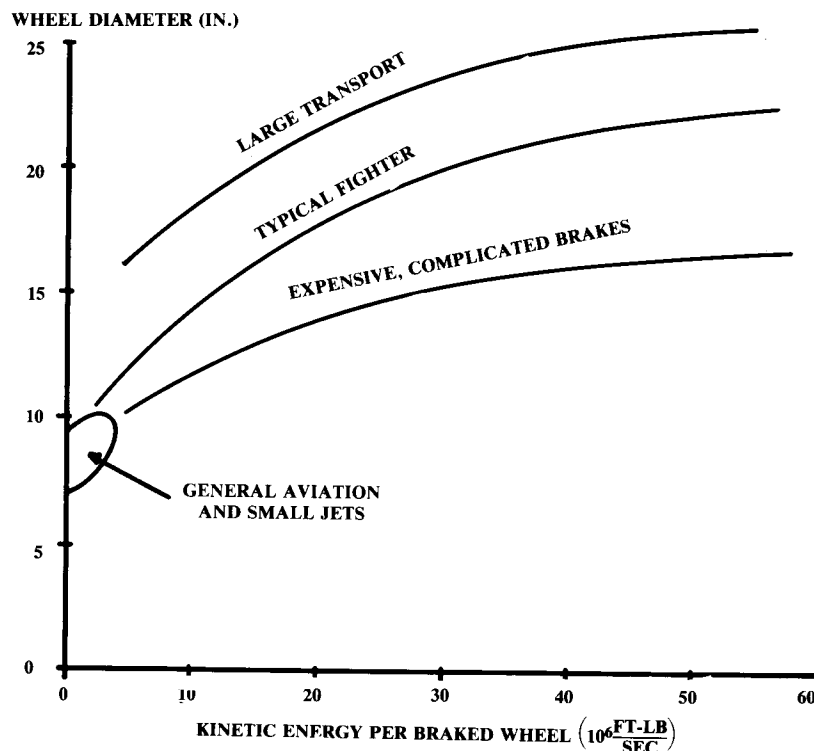


Fig. 11.7 Wheel diameter for braking.

it must be absorbed in the mass of the brake material. The amount of heat a brake can tolerate depends upon its size.

Figure 11.7 provides a statistical estimate of the required wheel rim diameter to provide a brake that can absorb a given amount of kinetic energy. Note that if an aircraft is initially designed with inadequate brakes, additional braking ability can be obtained in the same-sized wheels but only at a much higher cost by using exotic materials and complex design.

If the wheel rim diameter as estimated from Fig. 11.7 is larger than the rim diameter of the selected tire, a larger tire must be used. Alternatively, a brake which protrudes laterally from the tire can be used but will require a larger wheel well.

11.4 SHOCK ABSORBERS

Shock-Absorber Types

The landing gear must absorb the shock of a bad landing and smooth out the ride when taxiing. The more common forms of shock absorber are shown in Fig. 11.8.

The tires themselves provide some shock-absorbing ability by deflecting when a bump is encountered. Sailplanes and a few homebuilt aircraft have been built with rigid axles, relying solely upon the tires for shock absorbing.

Many World War I fighters used a rigid axle mounted with some vertical movement. The axle was attached to the aircraft with strong rubber chords

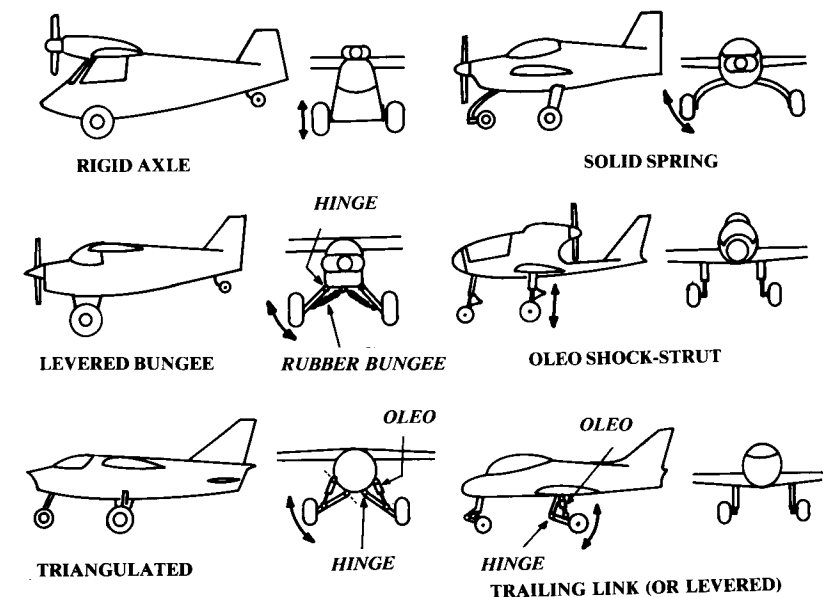


Fig. 11.8 Gear/shock arrangements.

("bungees") that stretched as the axle moved upward. This is rarely seen today.

The solid spring gear is used in many general-aviation aircraft (especially Cessna products). The solid spring is as simple as possible, but is slightly heavier than other types of gear.

Note that the solid-spring gear deflects with some lateral motion instead of straight up and down. This lateral motion tends to scrub the tires sideways against the runway, wearing them out. The solid spring has no damping other than this scrubbing action. The aircraft thus tends to bounce a lot, much like a car with bad shock-absorbers.

The levered bungee-chord gear was very common in early light aircraft such as the Piper Cub. The gear leg is pivoted at the fuselage. Rubber bungee chords underneath the gear are stretched as the gear deflects upward and outwards. This gear is light in weight but is high in drag. This gear also causes lateral scrubbing of the tires.

The oleopneumatic shock strut, or "oleo," is the most common type of shock-absorbing gear in use today (Fig. 11.9). The oleo concept was patented in 1915 as a recoil device for large cannons. The oleo combines a spring effect using compressed air with a damping effect using a piston which forces oil through a small hole (orifice). For maximum efficiency, many oleos have a mechanism for varying the size of the orifice as the oleo compresses ("metered orifice").

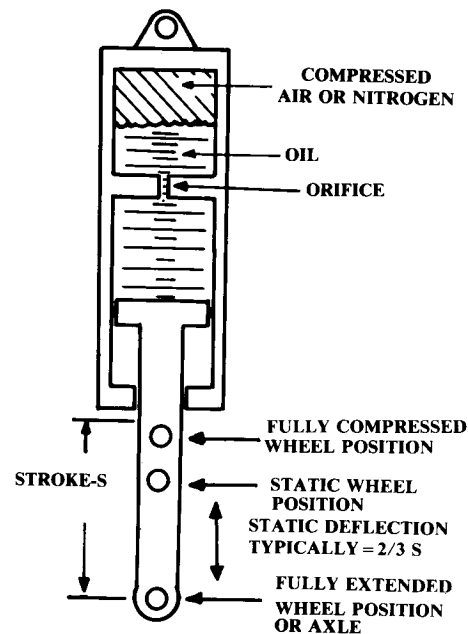


Fig. 11.9 Oleo shock absorber (most simple type).

When used as a shock-strut, the oleo itself must provide the full required amount of wheel deflection, which can lengthen the total landing-gear height. Also, the oleo strut must be strong enough to handle the lateral and braking loads of the wheels. To repair or replace the oleo strut, the entire wheel assembly must be removed because it is attached to the bottom of the strut.

The triangulated gear is similar to the levered bungee gear. When the triangulated gear is deflected, an oleopneumatic shock absorber is compressed. This provides a leveraged effect in which the oleo can be shorter than the required wheel travel. This is especially useful for carrier-based aircraft such as the A-7 that require large amounts of wheel travel to absorb the carrier-landing impact loads.

On a triangulated gear, the oleo can be replaced without removing the wheel assembly. The wheel lateral and braking loads are carried by the solid gear legs, which reduces the oleo weight. However, the complete triangulated gear is usually a little heavier than the oleo shock-strut gear. Also, there is a tire-scrubbing effect that shortens tire life.

The triangulated gear is sometimes seen on smaller aircraft using rubber blocks or springs in compression instead of an oleopneumatic shock absorber. The rubber blocks or springs can be inside the fuselage which streamlines the exposed part of the gear but requires the gear leg to support the aircraft's weight in a cantilevered fashion. This increases the gear weight.

The trailing-link, or levered, gear resembles the triangulated gear, but with the solid gear leg running aft rather than laterally. This gear is common for carrier-based aircraft such as the F-18 where it provides the large amounts of gear travel required for carrier landings. Typically the pivot point of the lower gear leg is slightly in front of the tire, less than one tire radius in front of the tire.

The levered gear allows the wheel to travel aft as it deflects. This is very desirable for operations on rough fields. If a large rock or other obstacle is encountered, the aft motion of the wheel gives it more time to ride over the obstacle. For this reason the levered gear was used on the North American Rockwell OV-10.

The triangulated gear and levered gear provide a mechanical lever effect that reduces the deflection of the shock-absorber oleo. However, this also increases the forces on the shock-absorber oleo, which increases its required diameter. The mechanical advantage of the triangulated or levered gear is determined from the actual dimensions of the gear layout, and is used to size the shock absorber.

Stroke Determination

The required deflection of the shock-absorbing system (the "stroke") depends upon the vertical velocity at touchdown, the shock-absorbing material, and the amount of wing lift still available after touchdown. As a rough rule-of-thumb, the stroke in inches approximately equals the vertical velocity at touchdown in (ft/s).

Table 11.4 Shock absorber efficiency

Type	Efficiency η
Steel leaf spring	0.50
Steel coil spring	0.62
Air spring	0.45
Rubber block	0.60
Rubber bungee	0.58
Oleopneumatic	
– Fixed orifice	0.65–0.80
– Metered orifice	0.75–0.90
Tire	0.47

The vertical velocity (or “sink speed”) at touchdown is established in various specifications for different types of aircraft. Most aircraft require 10 ft/s vertical velocity capability. This is substantially above the 4–5 ft/s that most passengers would consider a “bad” landing.

While most Air Force aircraft require only 10 ft/s, Air Force trainer aircraft require 13 ft/s. Due to steeper descent angles, Ref. 32 suggests 15 ft/s for short takeoff and landing (“STOL”) aircraft.

Carrier-based naval aircraft require 20 or more ft/s vertical velocity, which is much like a controlled crash! This is the reason that carrier-based aircraft tend to use triangulated or levered gear, which provide longer strokes than shock-strut gear.

In most cases it may be assumed that the wing is still creating lift equal to the aircraft’s weight during the time that the shock absorber is deflecting. The detailed shock-absorber calculations for FAR-23 aircraft must assume that only two-thirds of the aircraft’s weight is supported by the wing during touchdown. However, this can be ignored for initial stroke calculations.

The vertical energy of the aircraft, which must be absorbed during the landing, is defined in Eq. (11.8). This kinetic energy is absorbed by the work of deflecting the shock absorber and tire.

$$KE_{\text{vertical}} = \left(\frac{1}{2}\right) \left(\frac{W_{\text{landing}}}{g}\right) V_{\text{vertical}}^2 \quad (11.8)$$

where

W = total aircraft weight

g = 32.2 ft/s²

If the shock absorber were perfectly efficient, the energy absorbed by deflection would be simply the load times the deflection. Actual efficiencies of shock absorbers range from 0.5–0.9, as provided in Table 11.4. The actual energy absorbed by deflection is defined in Eq. (11.9).

$$KE_{\text{absorbed}} = \eta LS \quad (11.9)$$

where

η = shock-absorbing efficiency

L = average total load during deflection (not lift!)

S = stroke

For tires it is assumed that the tire deflects only to its rolling radius, so the “stroke” (S_T) of a tire is equal to half the diameter minus the rolling radius.

Combining Eqs. (11.8) and (11.9) and assuming that the shock absorber and tire both deflect to absorb the vertical kinetic energy yields:

$$\left(\frac{1}{2}\right) \left(\frac{W_{\text{landing}}}{g}\right) V_{\text{vertical}}^2 = (\eta LS)_{\text{shock absorber}} + (\eta_T L S_T)_{\text{tire}} \quad (11.10)$$

Note in Eq. (11.10) that the number of shock absorbers doesn’t enter into the equation. Remember that “ L ” is the average total load on the shock absorbers during deflection. The number of shock absorbers affects the diameter of the shock absorbers but not the required stroke.

The shock absorbers and tires act together to decelerate the aircraft from the landing vertical velocity to zero vertical velocity. The vertical deceleration rate is called the “gear load factor” (N_{gear}). Gear load factor is the average total load summed for all of the shock absorbers divided by the landing weight, and is assumed to be constant during touchdown. N_{gear} is defined in Eq. (11.11) and typically equals three.

$$N_{\text{gear}} = L/W_{\text{landing}} \quad (11.11)$$

The gear load factor determines how much load the gear passes to the airframe, which affects the airframe structural weight as well as crew and passenger comfort during the landing. Table 11.5 provides typical gear load factors permitted for various types of aircraft.

Substituting Eq. (11.11) into Eq. (11.10) yields Eq. (11.12) for shock-absorber stroke. Note that the equation for stroke does not include any terms containing the aircraft weight. For the same required landing vertical velocity and gear load-factor, an airliner and an ultralight would require the same stroke!

$$S = \frac{V_{\text{vertical}}^2}{2g \eta N_{\text{gear}}} - \frac{\eta_T}{\eta} S_T \quad (11.12)$$

Table 11.5 Gear load factors

Aircraft type	N_{gear}
Large bomber	2.0–3
Commercial	2.7–3
General aviation	3
Air Force fighter	3.0–4
Navy fighter	5.0–6

The stroke calculated by Eq. (11.12) should be increased by about 1 in. as a safety margin. Also, a stroke of 8 in. is usually considered a minimum, and at least 10–12 in. is desirable for most aircraft.

Nosewheel stroke is generally set equal to or slightly larger than main-wheel stroke to provide a smooth ride while taxiing.

Note that the stroke defined by Eq. (11.12) is a vertical distance. If a type of gear is used which produces some lateral motion of the wheel, the total distance the wheel moves must provide the required stroke in a vertical direction.

Equation (11.12) defines the total stroke required. If a levered or triangulated gear is used, the required stroke of the oleo, bungee, or rubber block is reduced and is determined by dividing the total stroke by the mechanical advantage. The actual load on the shock absorber is increased (multiplied) by the mechanical advantage.

Oleo Sizing

The actual dimensions of an oleo shock absorber or shock strut can now be estimated. The total oleo stroke is known. For most types of aircraft the static position is approximately 66% of the distance from the fully extended to the fully compressed position (see Fig. 11.9). For large transport aircraft, the static position is about 84% of stroke above the fully extended position. For a general-aviation aircraft the static position is typically about 60% of stroke above the extended position.

The total length of the oleo including the stroke distance and the fixed portion of the oleo will be approximately 2.5 times the stroke. For an aircraft with the desired gear attachment point close to the ground, this minimum oleo length may require going to a levered gear.

Oleo diameter is determined by the load carried by the oleo. The main wheel oleo load is the static load found from Eq. (11.1) divided by the number of main-wheel oleos (usually two). The nose-wheel oleo load is the sum of the static and dynamic loads [Eqs. (11.2) and (11.4)]. These loads must be increased by the mechanical advantage if levered or triangulated gear is used.

The oleo carries its load by the internal pressure of compressed air, applied across a piston. Typically an oleo has an internal pressure (“ P ”) of 1800 psi. Internal diameter is determined from the relationship which states that force equals pressure times area. The external diameter is typically 30% greater than the piston diameter, so the external oleo diameter can be approximated by Eq. (11.13).

$$D_{\text{oleo}} = 1.3 \sqrt{\frac{4L_{\text{oleo}}}{P\pi}} \cong 0.04 \sqrt{L_{\text{oleo}}} \quad (11.13)$$

where L_{oleo} = load on the oleo in pounds.

Solid-Spring Gear Sizing

Figure 11.10 illustrates the deflection geometry for a solid-spring gear leg. The total stroke as determined by Eq. (11.12) is the vertical component of

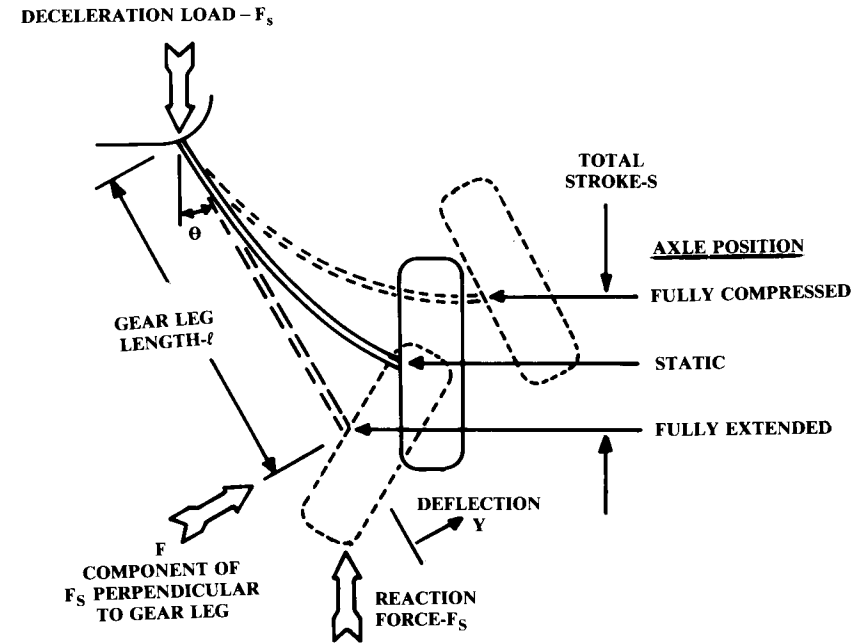


Fig. 11.10 Solid spring gear deflection.

the deflection of the gear leg. Note that the wheel is mounted so that it is vertical when the gear leg is deflected under the static load. This provides even tire wear.

If the gear leg is not excessively tapered, it may be approximated as a constant-cross-section bending beam using the average values of beam width (w) and thickness (t).

The load on the gear leg in the fully deflected position is the force required to produce the gear load factor, N_{gear} . Assuming there are two gear legs yields Eq. (11.14). The component of the load on the gear that is perpendicular to the gear leg is defined by Eq. (11.15).

$$F_s = WN_{\text{gear}}/2 \quad (11.14)$$

$$F = F_s(\sin\theta) \quad (11.15)$$

The deflection “ y ” perpendicular to the gear leg is related to the stroke by Eq. (11.16), and is calculated by the structural bending-beam equation [Eq. (11.17)—see Chapter 14]. Substituting Eqs. (11.14)–(11.16) into Eq. (11.17) yields the equation for the stroke “ S ” of a solid-spring gear leg:

$$S = y \sin\theta \quad (11.16)$$

$$y = Fl^3/3EI \quad (11.17)$$

$$S = F_s (\sin^2 \theta) \frac{l^3}{3EI} \quad (11.18)$$

where

I = beam's moment of inertia

E = material modulus of elasticity (psi)

(approximately 30 million for steel; 10 million for aluminum)

For a rectangular-cross-section gear leg, the moment of inertia is defined by Eq. (11.19).

$$I = \frac{wt^3}{12} \quad (11.19)$$

The static deflection is also determined using Eq. (11.18), but from the static wheel-load for the force (F_s).

11.5 CASTORING-WHEEL GEOMETRY

For ground steering, a nosewheel or tailwheel must be capable of being castored (turned). The castoring can introduce static and dynamic stability problems causing "wheel shimmy," a rapid side-to-side motion of the wheel that can tear the landing gear off the airplane.

Prevention of shimmy is accomplished by selection of the "rake angle" and "trail," as shown in Fig. 11.11. In some cases a frictional shimmy damper is also used to prevent shimmy. This can be a separate hydraulic plunger or simply a pivot with a lot of friction.

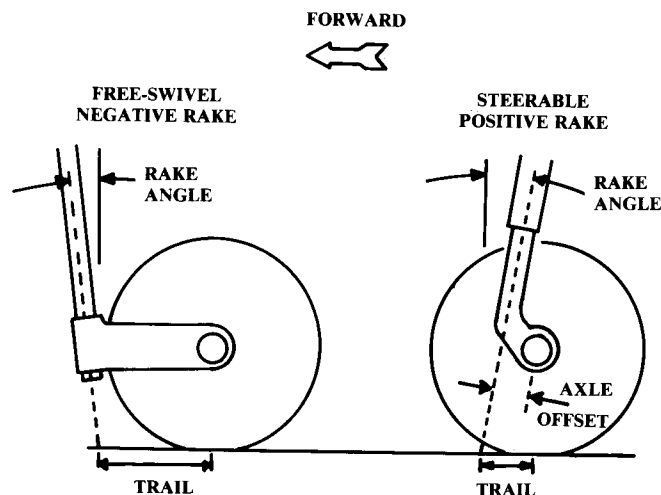


Fig. 11.11 Castoring wheel geometry.

If the castoring wheel is free to swivel, shimmy can be prevented by using a small negative angle of rake (4–6 deg), and trail equal to 0.2–1.2 times the tire radius (0.2 is typical for tailwheels). If the trail is less than the tire radius, a shimmy damper may be required.

If the nosewheel is free to swivel, the pilot must steer the aircraft on the ground using only the brakes. This increases brake wear and presents a great danger if one brake fails during takeoff or landing.

Note that tailwheels are always designed as if they are free to swivel. Steerable tailwheels are connected to the rudder pedals by soft springs that don't affect the wheel dynamics. Most larger tailwheel aircraft have provisions for locking the tailwheel during takeoff and landing.

For most tricycle-geared aircraft, a steering linkage is connected to the rudder pedals or a separate steering wheel, providing positive control of the turning angle. A key objective in the design of a steerable nosewheel is to reduce the required control forces while retaining dynamic stability.

This is done by minimizing the trail through use of a positive rake angle. Note that the weight of the aircraft tends to cause this gear to "flop over," i.e., the gear is statically unstable. This is prevented by the control linkages.

For a large aircraft with a steerable nose-wheel, the rake angle should be about 7 deg positive, and the trail should be at least 16% of the tire radius. For smaller aircraft, rake angles up to 15 deg and trail of about 20% are used.

11.6 GEAR-RETRACTION GEOMETRY

At this point, the required sizes for the wheels, tires, and shock absorbers are known, along with the required down locations of the wheels. The one remaining task is to find a "home for the gear" in the retracted position.

A poor location for the retracted gear can ruin an otherwise good design concept! A bad choice for the retracted position can chop up the aircraft structure (increasing weight), reduce the internal fuel volume, or create additional aerodynamic drag.

Figure 11.12 shows the options for main-landing-gear retracted positions. Locating the gear in the wing, in the fuselage, or in the wing-fuselage junction produces the smallest aerodynamic penalty but tends to chop up the structure. Gear in the wing reduces the size of the wing box, which increases weight and may reduce fuel volume. Gear in the fuselage or wing-fuselage junction may interfere with the longerons. However, the aerodynamic benefits of these arrangements outweigh the drawbacks for higher-speed aircraft.

Virtually all civilian jet transports retract the gear into the wing-fuselage junction. Most low-wing fighters retract the gear into the wing or wing-fuselage junction, while mid- and high-wing fighters retract the gear into the fuselage.

While some slower aircraft retract the gear into the wing, fuselage, or wing-fuselage junction, many retract the gear into the nacelles or a separate gear pod. This reduces weight significantly because the wing and fuselage structure is uninterrupted.

The wing-podded arrangement is rarely seen in Western aircraft designs (A-10), but is used in Soviet designs even for jet transports and bombers.

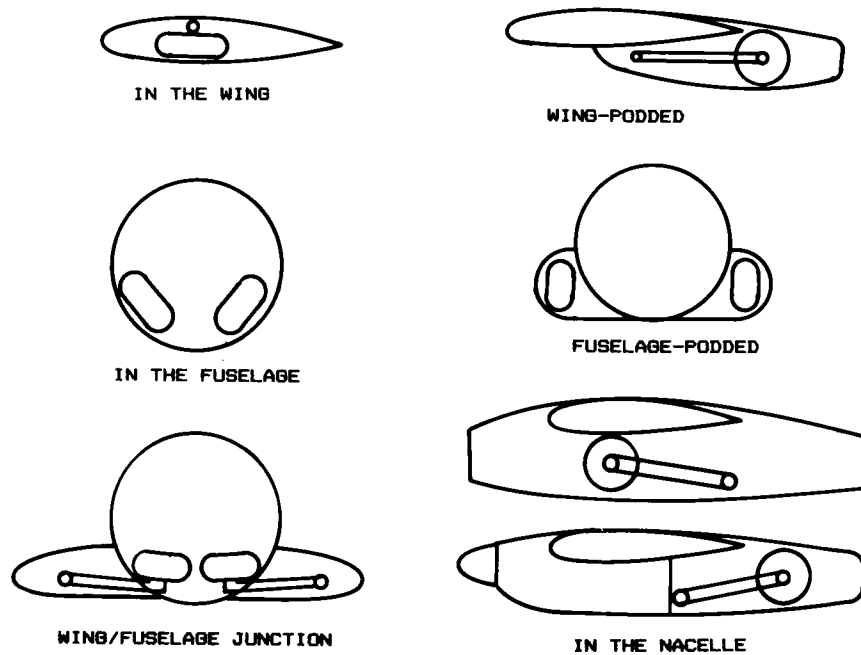


Fig. 11.12 "A home for the gear."

The aerodynamic penalty is minimized by placing the pods at the trailing edge of the wing, where some "area-ruling" benefit is obtained.

The fuselage-podded arrangement is common for high-winged military transports where the fuselage must remain open for cargo. The drag penalty of the pods can be substantial.

Retraction of the gear into the nacelles behind the engine is typical for propeller-driven aircraft. For jet-engined aircraft, nacelle-mounted landing gear must go alongside the engine, which widens the nacelle, increasing the drag.

Most mechanisms for landing-gear retraction are based upon the "four-bar linkage." This uses three members (the fourth bar being the aircraft structure) connected by pivots. The four-bar linkage provides a simple and lightweight gear because the loads pass through rigid members and simple pivots.

Several variations of four-bar linkage landing gear are shown in Fig. 11.13. The oldest form of four-bar linkage for landing gear retraction is shown in front view (Fig. 11.13a), where the wheel is at the bottom of a vertical gear member attached to parallel arms, which in turn attach to the fuselage. The gear retracts by pivoting the arms upward and inward. This was widely used during the 1930's, and is seen today in modified form on the MiG-23.

Figure 11.13b shows the typical retraction arrangement for nose wheels. The diagonal arm is called a "drag brace" because it withstands the aerody-

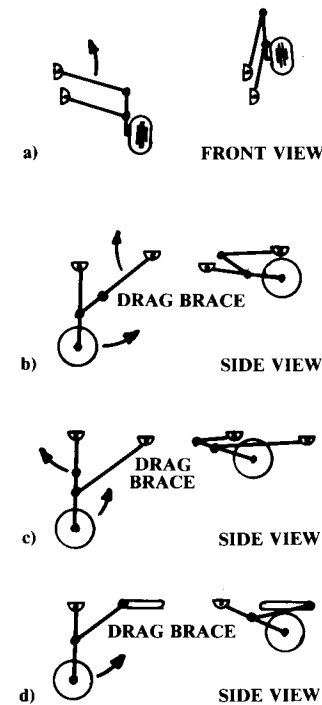


Fig. 11.13 Landing gear retraction.

amic loads (as well as braking loads). The drag brace breaks at the middle for retraction.

The drag brace may be behind the wheel with the gear retracting rearward or it may be in front of the wheel with the gear retracting forward. The latter is preferable because the air loads will blow the gear down in the event of a hydraulic failure.

In Fig. 11.13c the vertical gear member breaks for retraction instead of the drag brace. This has the advantage of reducing the length of the retracted gear, but is usually heavier. This gear was used on the DC-3 and several World War II bombers.

Figure 11.13d shows the use of a sliding pivot rather than a four-bar linkage. The sliding motion is frequently provided by a wormscrew mechanism that is rotated to retract the gear. This is usually heavier than a four-bar linkage because the entire length of the wormscrew must be strong enough for the landing-gear loads. However, this gear is very simple and compact.

The retraction concepts as shown in Fig. 11.13 are for nose or main wheels that retract in a fore or aft direction. However, the same basic concepts can be used for main wheels that retract inwards or outwards. These illustrations then become front views, and the tires are redrawn accordingly. The gear members labeled "drag brace" become "sway braces" because they provide lateral support for the gear in this arrangement.

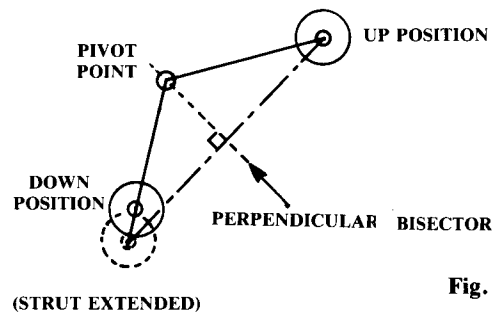


Fig. 11.14 Pivot point determination.

There are dozens of additional geometries for gear retraction based upon the four-bar linkage and other concepts. For these, see Ref. 32 or 33.

The landing-gear leg is attached to the aircraft at the “pivot point.” This is determined as shown in Fig. 11.14. The pivot point can lie anywhere along the perpendicular bisector to the line connecting the up and down positions of the wheel.

Normally the gear strut is allowed to extend fully before retraction, as shown in Fig. 11.14, although it is possible to install a “strut compressor” that causes the gear to be retracted with the strut in the compressed position. This should be used only when internal space is absolutely unavailable for the fully extended strut.

It is also possible to provide a “rotator” mechanism or “planing link” that will change the angle between the gear leg and the wheel axis when the gear is retracted. This is sometimes required to permit the wheel to lie flat inside the wheel well when retracted. This is fairly simple and is seen on many military aircraft such as the F-16. However, all such mechanisms should be avoided if possible due to the increased weight, complexity, and maintenance.

11.7 SEAPLANES

Seaplanes were important during the early days of aviation because of the limited number of good airports. Early commercial over-water flights were made exclusively by seaplanes, for safety. Most early speed records were held by seaplanes (or floatplanes) because the use of water for takeoff allowed long takeoff runs and hence high wing-loadings.

Today the seaplane concept is largely restricted to sportplanes, bush planes, and search-and-rescue aircraft. However, the notion of a “sea-sitter” strategic bomber which “loiters” for weeks at an unknown ocean location is revived every few years.

The hull of a seaplane and the pontoon of a float plane are based on the planing-hull concept. The bottom is fairly flat, allowing the aircraft to skim (plane) on top of the water at high speeds. A step breaks the suction on the after body. A vertical discontinuity, as shown in Fig. 11.15, the step can be straight in planview, or it can have an elliptical shape in planview to reduce aerodynamic drag.

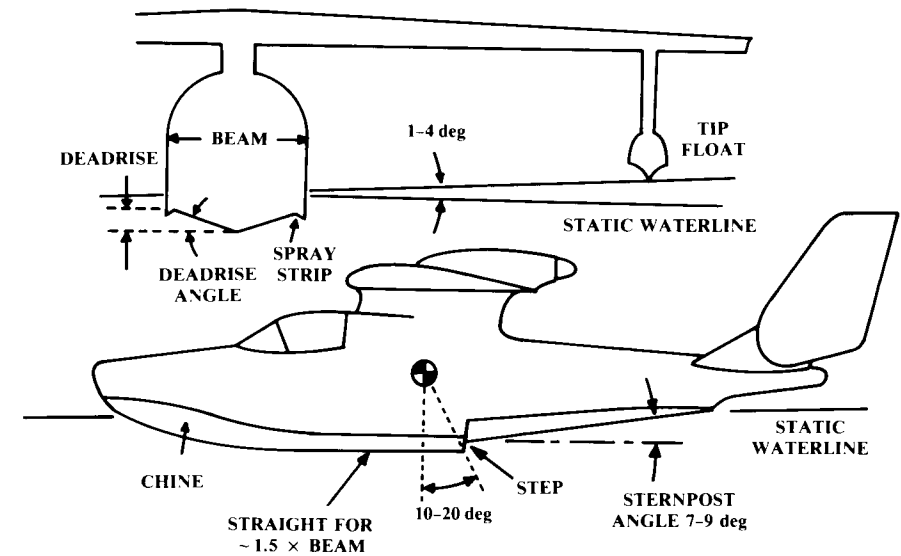


Fig. 11.15 Seaplane geometry.

While a few small seaplanes have been built with flat bottoms, most use a V-shaped bottom to reduce the water-impact loads. The height of the V is called the “deadrise,” and the angle is the “deadrise angle.” Deadrise angle must be increased for higher landing speeds, and should roughly follow Eq. (11.20). Deadrise angle is increased toward the nose to about 30–40 deg to better cut through waves.

$$\alpha_{\text{deadrise}} \cong \frac{V - 20}{2}, \text{ deg} \quad (11.20)$$

where V = stall speed in miles per hour.

To reduce water spray, spray strips can be attached to the edges of the bottom, as shown. These are angled about 30 deg below the horizon.

The ratio between the waterline length and “beam” (width) has a strong effect upon water resistance and landing impact. A wider hull has a lower water resistance due to its better planing ability but suffers a higher landing impact. Length-to-beam ratios vary from about six for a small seaplane to about 15 for a large one.

The step height should be about 5% of the beam. The step should be located on an angle about 10–20 deg behind the c.g. The bottom of the hull forward of the step should not be curved for a distance about equal to 1.5 times the beam. This is to reduce porpoising tendencies. Also, the hull bottom aft of the step (the “sternpost”) should angle upward about 8 deg.

For a true “flying boat” (i.e., seaplane with a boat-like fuselage), lateral stability on the water is usually provided by wing-mounted pontoons. These

should be located such that they contact the water when the aircraft tips sideways about 1 deg.

Determination of the static waterline is done using a modification of the fuselage-volume plotting technique previously described. First, a static waterline is assumed and drawn on the configuration layout. Then a volume plot is prepared from the cross-sectional areas of only those parts of the fuselage which are below the assumed waterline.

The area under the curve on the volume plot defines the fuselage submerged volume (i.e., below the waterline). This is multiplied by the density of water (62.4 lb/ft³) to determine the weight of aircraft which can be supported by that amount of displaced water.

The centroid of the area on the volume plot is the center of buoyancy, which should coincide with the c.g.. If either the weight of aircraft supported or the center of buoyancy are incorrect, another waterline must be assumed and a new volume plot prepared.

Water-resistance drag is very difficult to estimate. It depends upon the mechanics of wave production, and can vary widely for similar hull shapes. Also, water-resistance drag varies with speed. A seaplane hull can have a maximum water resistance at "hump speed" equal to 20% or more of the aircraft's weight.

Seaplane design and analysis should be based upon some published set of test results for a known hull shape. These can be found in early NACA reports. The major U.S. facility for testing seaplane and ship hulls is the Naval Ship Research and Development Center (NSRDC) near Washington, D.C.

For a rough estimate of a seaplane's takeoff distance, it can be assumed that the water-resistance drag will average about 10–15% of the waterborne weight. This is analogous to a rolling-friction coefficient (μ) of 0.10–0.15 in the takeoff calculations provided in Chapter 17.

11.8 SUBSYSTEMS

Aircraft subsystems include the hydraulic, electrical, pneumatic, and auxiliary/emergency power systems. Also, the avionics can be considered a subsystem (although to the avionics engineers, the airframe is merely the "mobility subsystem" of their avionics package!)

In general, the subsystems do not have a major impact on the initial design layout. However, later in the design cycle the configuration designer will have to accommodate the needs of the various subsystems, so a brief introduction is provided below. No attempt is made to provide examples or rules of thumb because the subsystems hardware varies widely between different classes of aircraft. Reference 11 provides additional information on subsystems.

Hydraulics

A simplified hydraulic system is shown in Fig. 11.16. Hydraulic fluid, a light oil-like liquid, is pumped up to some specified pressure and stored in an "accumulator" (simply a holding tank).

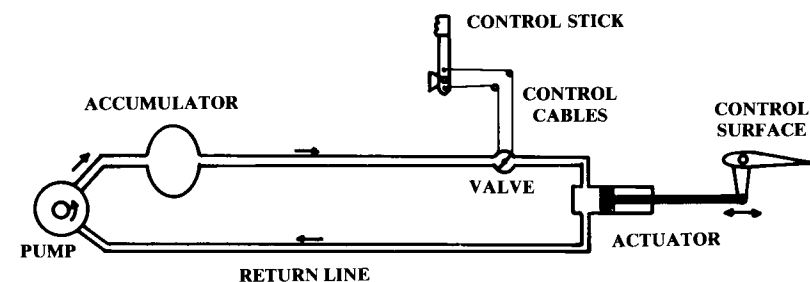


Fig. 11.16 Simplified hydraulic system.

When the valve is opened, the hydraulic fluid flows into the actuator where it presses against the piston, causing it to move and in turn moving the control surface. To move the control surface the other direction, an additional valve (not shown) admits hydraulic fluid to the back side of the piston. The hydraulic fluid returns to the pump by a return line.

To obtain rapid response, the valve must be very close to the actuator. The valve therefore cannot be in or near the cockpit, and instead is usually attached to the actuator.

In most current designs the pilot's control inputs are mechanically carried to the actuator by steel cables strung from the control wheel or rudder pedals to the valves on the actuators. In many new aircraft the pilot's inputs are carried electronically to electromechanical valves ("fly-by-wire").

Hydraulics are used for aircraft flight control as well as actuation of the flaps, landing gear, spoilers, speed brakes, and weapon bays. Flight-control hydraulic systems must also include some means of providing the proper control "feel" to the pilot. For example, the controls should become stiffer at higher speeds, and should become heavier in a tight, high-g turn. Such "feel" is provided by a combination of springs, bobweights, and air bellows.

In most cases the hydraulic system will impact the aircraft conceptual design only in the provision of space for the hydraulic pumps, which are usually attached to the engines. These should be copied from a similar aircraft if better information is not available.

Electrical System

An aircraft electrical system provides electrical power to the avionics, hydraulics, environmental-control, lighting, and other subsystems. The electrical system consists of batteries, generators, transformer-rectifiers ("TR's"), electrical controls, circuit breakers, and cables.

Aircraft generators usually produce alternating current (AC) and are located on or near the engines. TR's are used to convert the alternating current to direct current (DC). Aircraft batteries can be large and heavy if they are used as the only power source for starting.

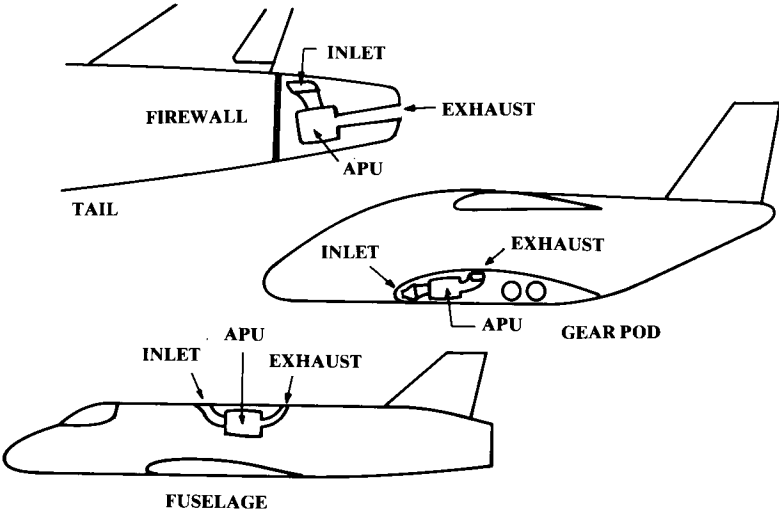


Fig. 11.17 APU installation.

Pneumatic System

The pneumatic system provides compressed air for pressurization, environmental control, anti-icing, and in some cases engine starting. Typically the pneumatic system uses pressurized air bled from the engine compressor. This compressed air is cooled through a heat exchanger using outside air. This cooling air is taken from a flush inlet inside the inlet duct (i.e., inlet secondary airflow) or from a separate inlet usually located on the fuselage or at the front of the inlet boundary-layer diverter.

The cooled compressor air is then used for cockpit pressurization and avionics cooling. For anti-icing, the compressor bleed air goes uncooled through ducts to the wing leading edge, inlet cowls, and windshield.

Compressed air is sometimes used for starting other engines after one engine has been started by battery. Also, some military aircraft use a ground power cart that provides compressed air through a hose to start the engine.

Auxiliary/Emergency Power

Large or high-speed aircraft are completely dependent upon the hydraulic system for flight control. If the hydraulic pumps stop producing pressure for any reason, the aircraft will be uncontrollable. If the pumps are driven off the engines, an engine flame-out will cause an immediate loss of control. For this reason, some form of emergency hydraulic power is required. Also, electrical power must be retained until the engines can be restarted. The three major forms of emergency power are the ram-air turbine (RAT), monopropellant emergency power unit (EPU), and jet-fuel EPU.

The ram-air turbine is a windmill extended into the slipstream. Alternatively, a small inlet duct can open to admit air into a turbine.

The monopropellant EPU uses a monopropellant fuel such as hydrazine to drive a turbine. The available monopropellants are all toxic and caustic, so monopropellant EPU's are undesirable for operational considerations. However, they have the advantage of not requiring any inlet ducts and can be relied upon to provide immediate power regardless of aircraft altitude, velocity, or attitude. Monopropellant EPU's must be located such that a small fuel leak will not allow the caustic fuel to puddle in the aircraft structure, possibly dissolving it!

Jet-fuel EPU's are small jet engines that drive a turbine to produce emergency power. These may also be used to start the main engines ("jet-fuel starter"). While they do not require a separate and dangerous fuel, the jet-fuel EPU's require their own inlet duct.

Most commercial transports and an increasing number of military aircraft use a jet-fuel auxiliary power unit ("APU"). An APU is much like an EPU but is designed and installed to allow continuous operation.

Usually an APU is designed to provide ground power for air conditioning, cabin lighting, and engine starting. This frees the aircraft from any dependence upon ground power carts. The APU is also used for in-flight emergency power, and in some cases is run continuously in-flight for additional hydraulic and/or electrical power.

The APU is actually another jet engine, and its installation must receive attention in the earliest design layout. The APU requires its own inlet and exhaust ducts, and must be contained in a firewalled structure. APU's have fairly high maintenance requirements so access is important.

To avoid high levels of noise, the inlet and exhaust of an APU should be directed upward. For in-flight operation of the APU, the inlet should ideally be in a high-pressure area and the exhaust in a low-pressure area. Also, the inlet should not be located where the exhaust of the jet engines or APU can be ingested. The exhaust of an APU is hot and noisy, and should not impinge upon aircraft structure or ground personnel.

Transport aircraft usually have the APU in the tail, as shown in Fig. 11.17. This removes the APU from the vicinity of the passenger compart

Table 11.6 Avionics weights

	Typical values
	$\frac{W_{\text{avionics}}}{W_{\text{empty}}}$
General aviation—single engine	.01–.03
Light twin	.02–.04
Turboprop transport	.02–.04
Business jet	.04–.05
Jet transport	.01–.02
Fighters	.03–.08
Bombers	.06–.08
Jet trainers	.03–.04

ment to reduce noise. The APU firewall is of minimum size, and the APU is easily accessible from a workstand.

Military transports with the landing gear in fuselage-mounted pods can place the APU in the pod. This provides ground-level access to the APU, but requires increased firewall area.

Fighters usually have the APU in the fuselage, near the hydraulic pumps and generators. This requires a firewall that completely encloses the APU.

APU installation is discussed in detail in Ref. 34.

Avionics

Avionics (a contraction of “aviation electronics”) includes radios, flight instruments, navigational aids, flight control computers, radar, and other aircraft sensors such as infrared detectors. For initial layout, it is necessary to provide sufficient volume in the avionics bays. Also, the nose of the aircraft should be designed to hold the radar.

On the average, avionics has a density of about 30–45 lb/ft³. The required avionics weight can be estimated from the aircraft empty weight (W_e), which is known at this point. Table 11.6 provides ratios between avionics weight and aircraft empty weight.

Estimation of radar size is very complex, and depends upon the desired detection range, radar cross section of the threat aircraft, and radar frequency. For initial design, the radar from a similar design should be used until the avionics group provides a definition of the required radar.

In the absence of better information, it can be assumed that a bomber will use a 40-in. radar, a large fighter will use a 35-in. radar, and a small fighter will use a 22-in. radar. Transport-aircraft radars are only for weather avoidance. They are small relative to the size of the aircraft nose and can be initially ignored.

12.1 INTRODUCTION

The previous chapters have presented methods for the design layout of a credible aircraft configuration. Initial sizing, wing geometry, engine installation, tail geometry, fuselage internal arrangement, and numerous other design topics have been discussed.

The initial sizing was based upon rough estimates of the aircraft’s aerodynamics, weights, and propulsion characteristics. At that time we could not calculate the actual characteristics of the design because the aircraft had not been designed yet!

Now the aircraft design can be analyzed “as-drawn” to see if it actually meets the required mission range. If not, we will resize the aircraft until it does.

Also, a variety of trade studies can now be performed to determine the best combination of design parameters (T/W , W/S , aspect ratio, etc.) to meet the given mission and performance requirements at the minimum weight and cost.

Additional analysis on the as-drawn aircraft is also required at this time to insure that stability and control requirements are met. In previous chapters, an approximate tail volume coefficient method was used for tail sizing. Now that the aircraft is drawn, we can analytically determine if the selected tail sizes are adequate.

These methods will be presented in Chapters 12–19. The overall objective of these chapters is the understanding of the sizing optimization and trade study process, not the presentation of particular theories and analysis methods. Presumably the student has spent several years studying aerodynamics, controls, structures, and propulsion.

The analysis techniques presented in these chapters, all approximate methods, illustrate the major parameters to be determined and provide realistic trends for trade studies. In many cases these are not the methods employed by the major aircraft companies, whose methods are highly computerized and cannot be presented in any single textbook. Also, each company uses many proprietary methods that are simply unavailable to students.

By using the simplified methods provided in this book, the student designer will experience the interaction of the major design variables but will not devote excessive time to the analytical tasks. This should leave more time for learning the basic principles of sizing optimization and trade studies, which are the same regardless of analytical techniques.